Transferable Mangrove Extent Mapping from Tanager-1 Hyperspectral Imagery Using Adaptive Spectral Thresholds and Pseudo-Labels

Using per-scene adaptive spectral thresholds for zero-shot transfer across four continents

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Mangrove store outsized blue carbon, yet resist monitoring at scale

Tree obstacles stand between the sensor and a reliable, transferable map.

1

Disproportionate blue-carbon stocks

Mangroves lock away far more carbon per hectare than most forests, making their extent a high value climate variable.

2

A spectral resolution gap

Conventional multispectral sensors cannot reliably separate mangrove from adjacent coastal vegetation across diverse sites.

3

Fixed thresholds do not travel

Index thresholds calibrated at one location routinely fail elsewhere (Munawaroh et al., 2025; Nie et a., 2026; Lassalle et al., 2023).

*Pic: P Mantha (Unsplash)

CORE IDEA

Calibrate spectral thresholds from each scene's own histogram, so no threshold, band, or label is imported from the training site. This is what enables reference-free, zero-shot transfer.

REIP and EMI use continuous red-edge and SWIR coverage that multispectral sensors lack (Lassalle 2023).

Hyperspectral bands resolve mangrove canopy structure finely enough for species-level work (Kamal & Phinn 2011; Yang 2024).

426

contiguous bands make per-scene calibration possible

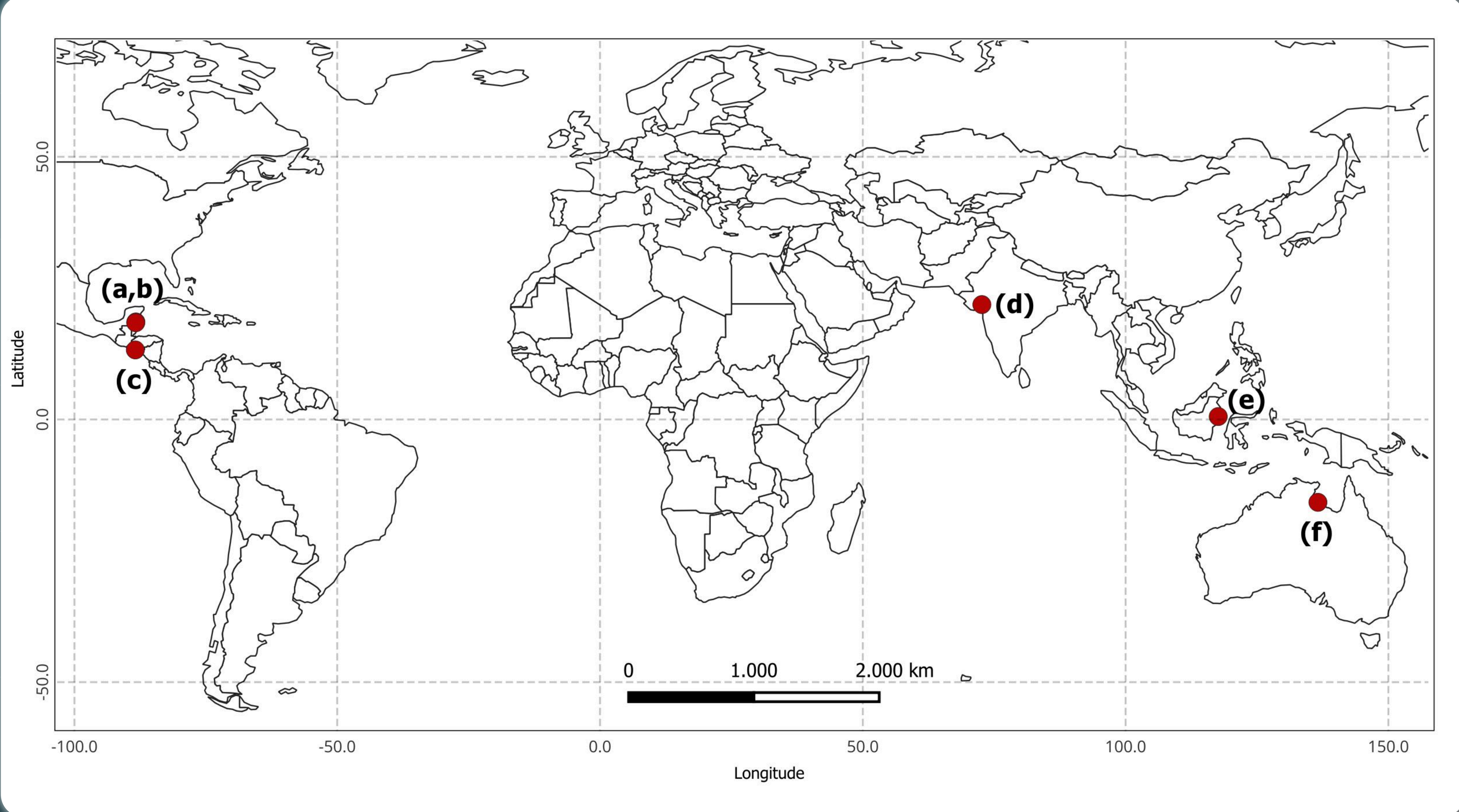
380-2500 nm
VISIBLE-SWIR RANGE

30 m
SPATIAL RESOLUTION

6 / 4
SCENES / CONTINENTS

SIX SCENES, FOUR CONTINENTS, ONE TRAINING ANCHOR

Trained on one Indonesian scene, transferred to four continents



(a,b) Belize 1 & 2; (c) El Salvador; (d) Gujarat, India; (e) Sangatta, Indonesia; (f) Australia

SITE	COASTAL SETTING	GMW V3 (HA)	ROLE
Sangatta	Tropical estuarine	1,216	**Training anchor**
Australia	Wet-dry estuarine	5,798	Transfer
Belize 1	Barrier reef	2,730	Transfer
Belize 2	Barrier reef	719	Transfer
El Salvador	Barrier reef	8,893	Transfer
Gujarat, India	Semi-arid intertidal	345	Transfer

Sangatta (Indonesia) is the sole training scene; the classifier is applied to the other five without retraining.

380-2500 nm
VISIBLE-SWIR RANGE

30 m
SPATIAL RESOLUTION

6 / 4
SCENES / CONTINENTS

1 TRAINED → **5** ZERO-SHOT

A deterministic, seven-stage pipeline

PRE-PROCESSING

Tanager-1
426 bands, 6 scenes; surface reflectance

Feature extraction
8 spectral indices + REIP, per scene

Coastal candidate mask
MNDWI + SAVI, Otsu <= 500m

MODEL DEVELOPMENT

Adaptive threshold calibration
Sangatta, bimodal + Otsu (MVI)

Pseudo-label generation
MVI AND NDMI

Model training
XGBoost, 5-fold CV, hyperparameter tuning

GMW v3 (2020)
Mangrove extent as independent reference

TRANSFER & VALIDATION

Inference model to all 6 scenes
per-scene recalibration, no retraining

Accuracy assessment
vs GMW v3; Kappa, F1, IoU

Outputs
Mangrove extent + commission-omission + table

Nine spectral features, two of them hyperspectral-only

NDVI
(NIR − Red)/(NIR + Red)
Rouse (1974); Huete (1988)

MNDWI
(Green − SWIR1)/(Green + SWIR1)
Xu (2006)

NDMI
(NIR − SWIR1)/(NIR + SWIR1)
Gao (1996)

CMRI
NDVI - MNDWI
Gupta (2018)

NDRE
(NIR − RedEdge)/(NIR + RedEdge)
Gitelson & Merzlyak (1994)

SAVI
(1+L)(NIR − Red)/(NIR + Red + L)
Huete (1988)

MVI
(NIR − Green)/(SWIR1 − Green)
Baloloy (2020)

HYPERSPECTRAL

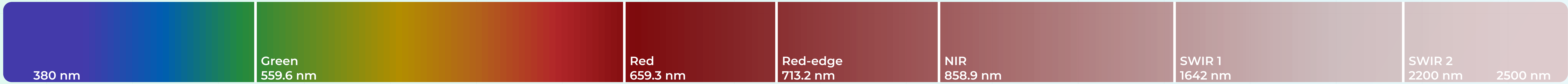
EMI
(NIR − SWIR2)/(NIR + SWIR2)
Rahmila (2026)

HYPERSPECTRAL

REIP
red-edge inflection, 700–740 nm
Guyot (1988); Cho & Skidmore (2006)

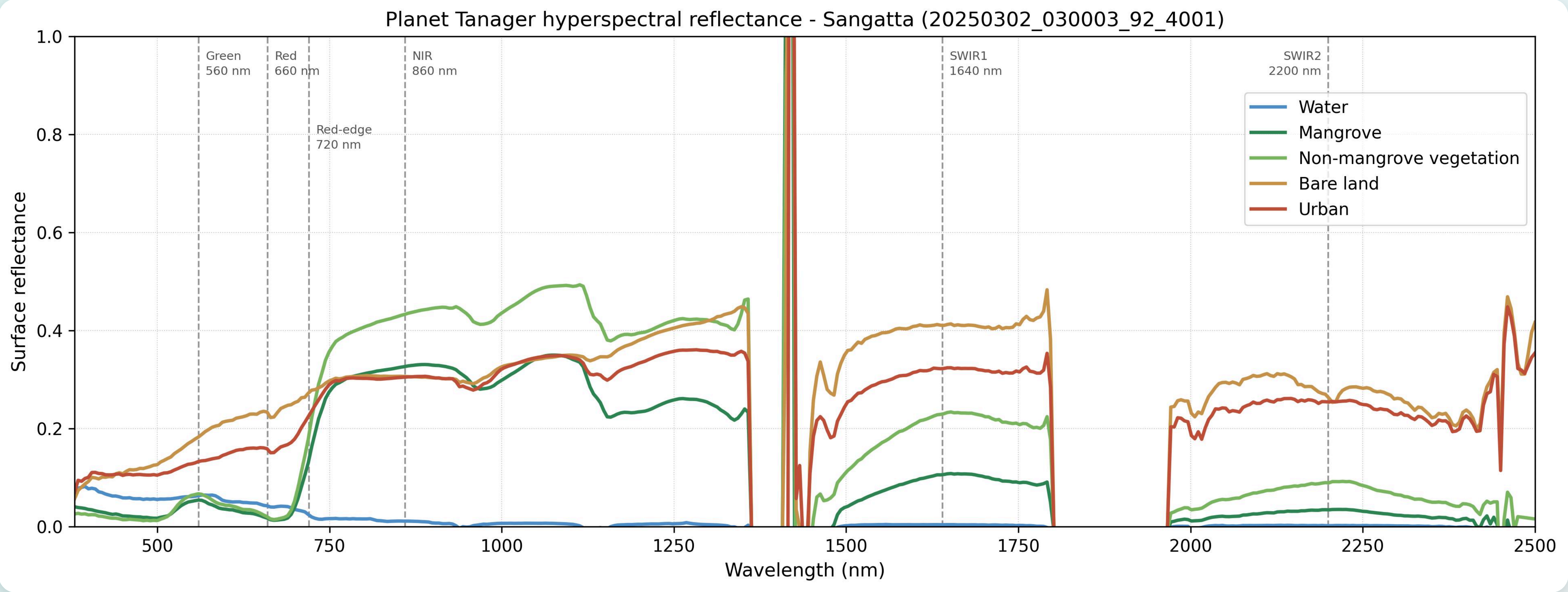
Six Tanager bands (559.6 to 2200.7 nm) feed all nine features; EMI and REIP use continuous red-edge and SWIR coverage unavailable to multispectral sensors (Lassalle 2023; Kamal & Phinn 2011).

Six bands, matched to each index's reference wavelength



BAND	REFERENCE (NM)	NEAREST TANAGER BAND (NM)	FEEDS	BAND-POSITION REFERENCE
Green	560	559.6	MNDW, MVI	Xu (2006)
Red	660	659.3	NDBI, SAVI, CMRI	Huete (1998)
Red-edge	720	719.2	NDRE, REIP	Gitelson & Merzlyak (1994)
NIR	860	858.9	NDVI, NDMI, SAVI, CMRI, NDRE, EMI, MVI	Gao (1996), Huete (1988)
SWIR 1	1640	1642	MNDWI, NDMI, MVI	Xu (2006), Gao (1996)
SWIR 2	2200	2200	EMI	Rahmila (2026)

Tanager reconstructs 426 bands across 380 to 2500 nm; for each index the band nearest its reference wavelength is selected within a 5 nm tolerance. [src: preprocessing.py - RELEVANT_WAVELENGTHS]



The spikes near 1400 nm are atmospheric water-vapor absorption bands, where surface signal drops to near-zero. This is standard in all VIS-SWIR hyperspectral data (EMIT, PRISMA, AVIRIS); these bands are masked before analysis. The six bands we use for indices all sit in clean atmospheric windows.

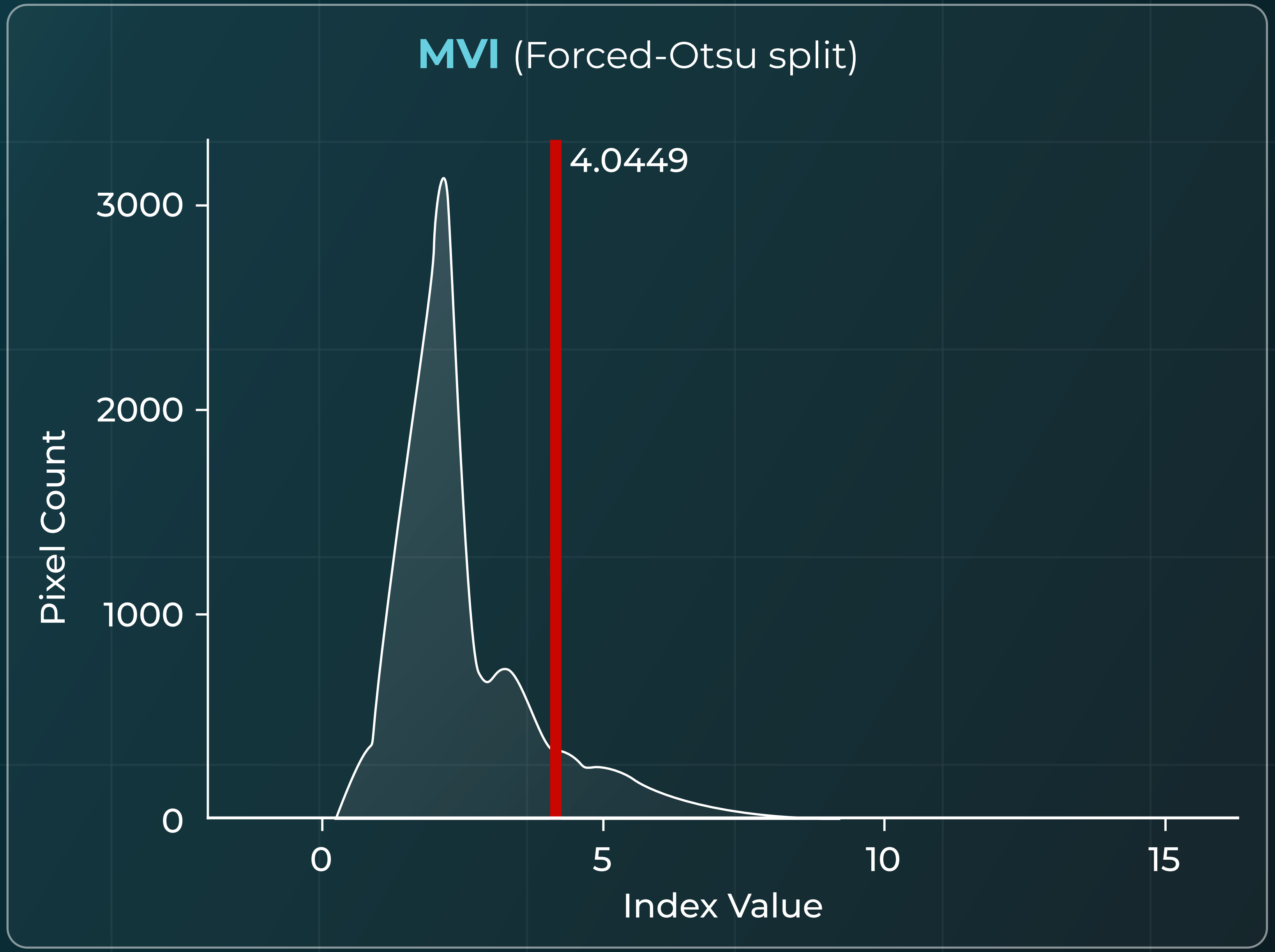
Each scene calibrates its own thresholds

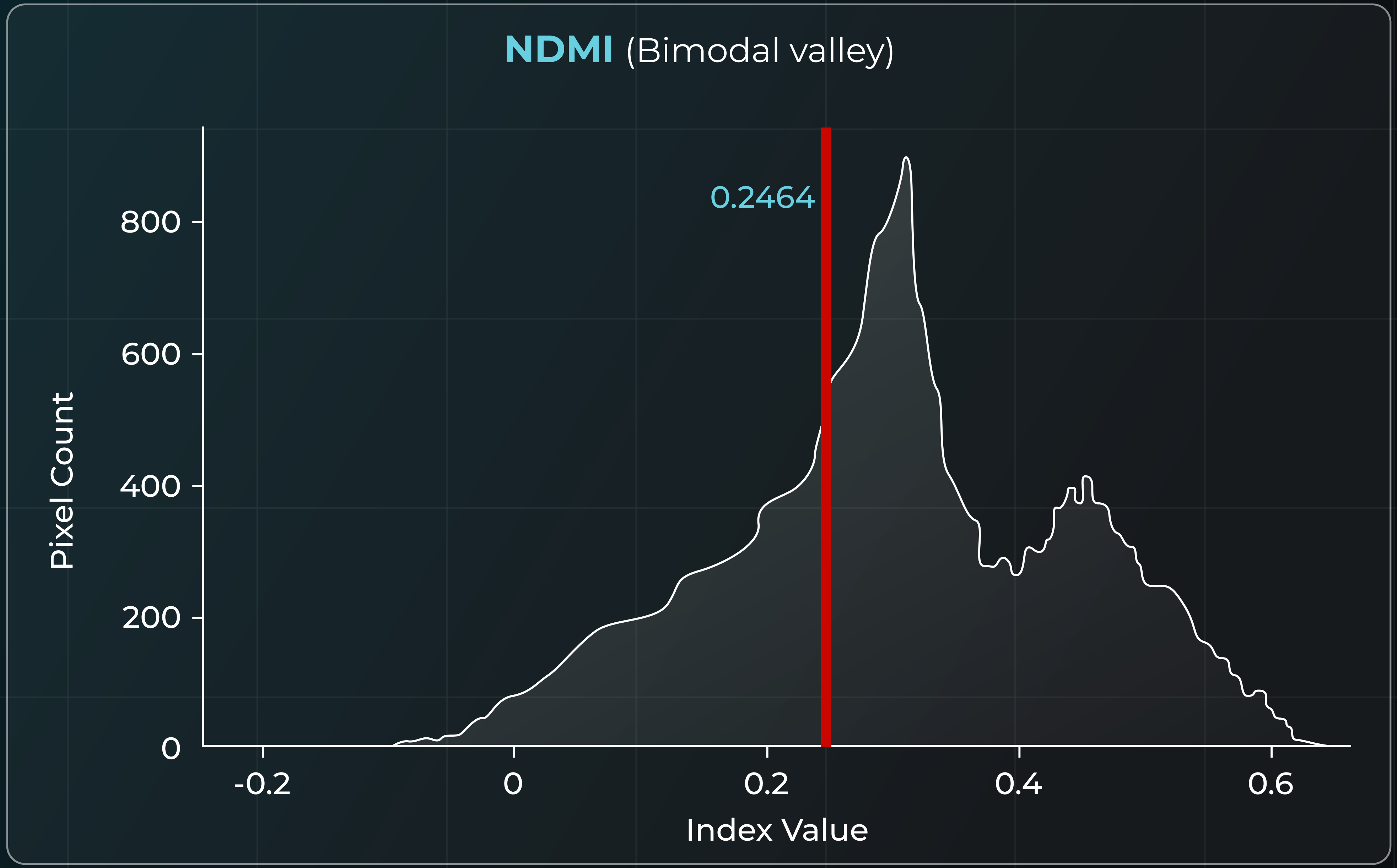
Adaptive Thresholds: Sangatta (candidate zone, n=57,014 px)

57.014 px
CANDIDATE ZONE (N)

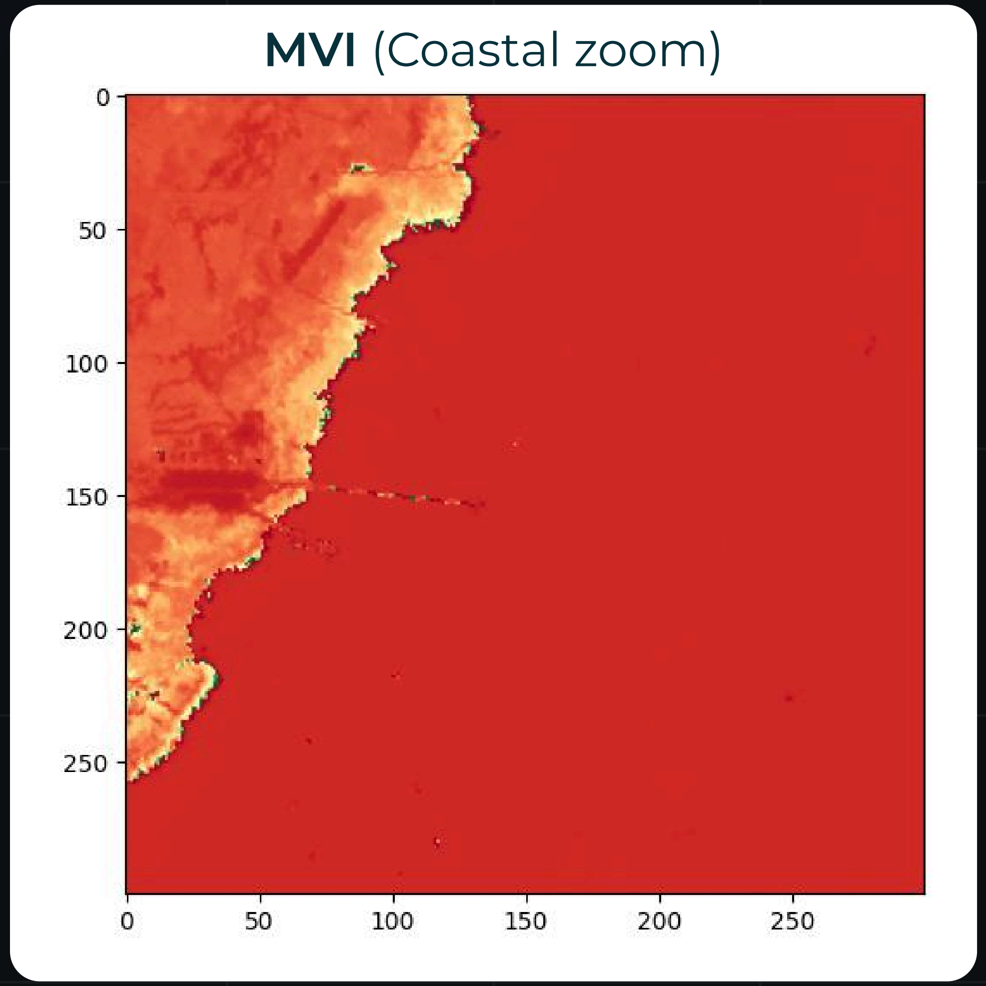
4.0449
MVI THRESHOLD

0.2464
NDMI THRESHOLD





Per-scene MVI (forced-Otsu) and NDMI (bimodal valley), Sangatta candidate zone. [src: NB01 · thresholds_sangatta.png]



MVI coastal zoom: the spike pixels are confined to the open-water body (right side), not the vegetated coast, confirming the artifact is a water divide-by-near-zero effect rather than a land-cover signal.

- MVI = (NIR − Green) / (SWIR1 − Green): a difference ratio, not bounded to [-1, 1]
- Spike near 20 = divide-by-near-zero where SWIR1 ≈ Green (open water), then capped
- Far right of the threshold (4.0449) and confined to water: classification unaffected

The candidate zone is hard-capped at 500 m from water. Without the cap the Otsu buffer expanded to ~1,350 m and admitted inland (non-mangrove vegetation).

Because calibration is per-scene, nothing is imported from the training site.

Thresholds auto-generate labels; XGBoost learns the rests

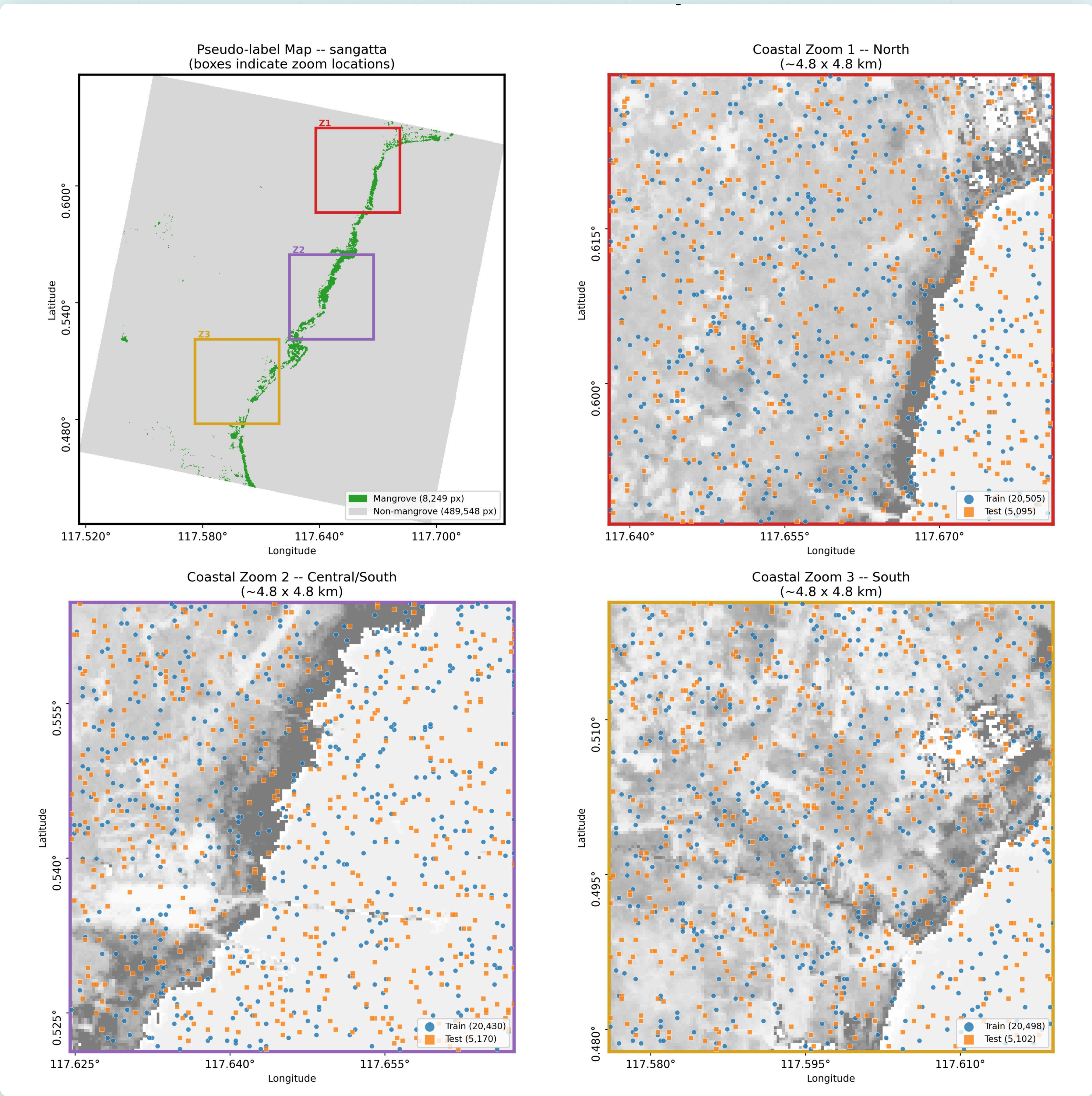
1. **AND-logic pseudo-labels**
   Pixels above both MVI and NDMI thresholds become positives: 8,249 mangrove within the candidate zone.

2. **Whole-scene training matrix**
   497,797 pixels, split 80/20 (398,237 train / 99,560 test); XGBoost tuned by RandomizedSearchCV (50 iter, 5-fold, F1).

KEY INSIGHT

0.962 CV-F1 vs **0.615** GMW-F1

The ceiling is pseudo-label quality, not model capacity. Effort should go to label refinement, not more tuning.



[src: NB02 - pseudo_label_map_sangatta_20250302_030003_92_4001.png]

Training-site accuracy against GMW v3 (2020)

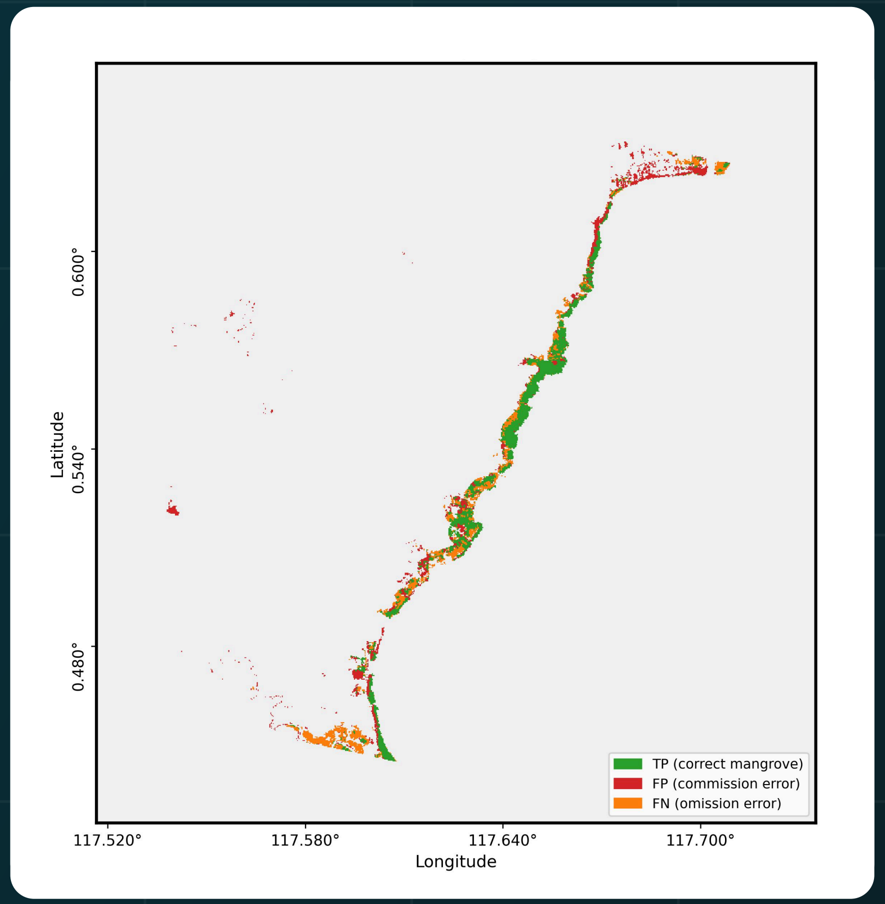
0.887
OVERALL ACCURACY

0.548
COHEN'S KAPPA

0.615
F1 (MANGROVE)

0.444
IOU

GMW v3 (2020) is independent and never used in training. The 2020-vs-2025 mismatch scores five years of change as error, making Precision and Kappa conservative.

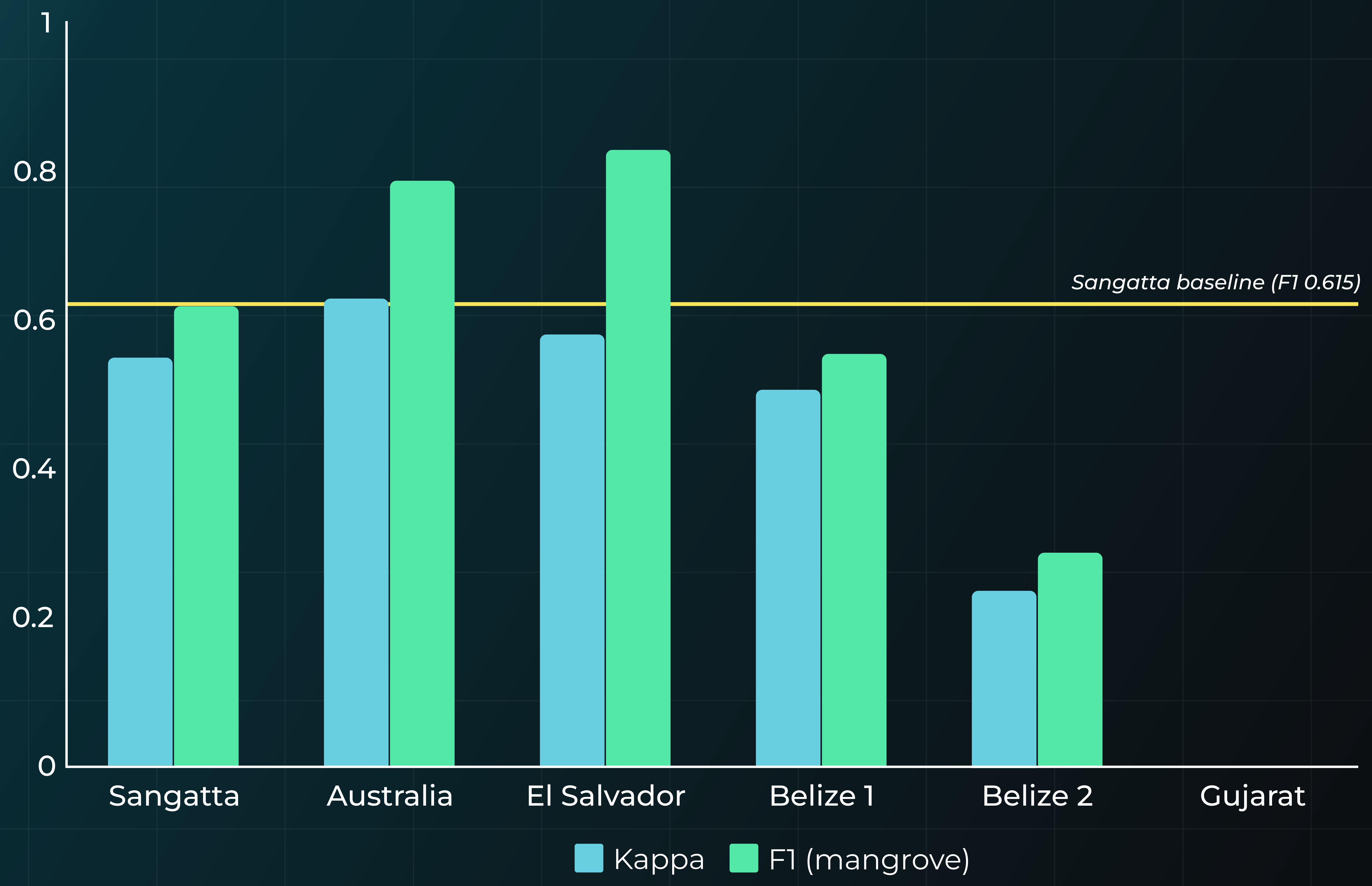


Sangatta commission-omission error map.

	Ref: Mangrove	Ref: Non-mangrove
Pred: Mangrove	5.146 TP	3.091 FP
Pred: Non-mangrove	3.360 FN	45.417 TN

Confusion matrix, XGBoost Tuned vs GMW v3, candidate zone (57,014 px). [src: memo Table 2 - NB02 gmw_eval CSV]

One model, applied across four continents



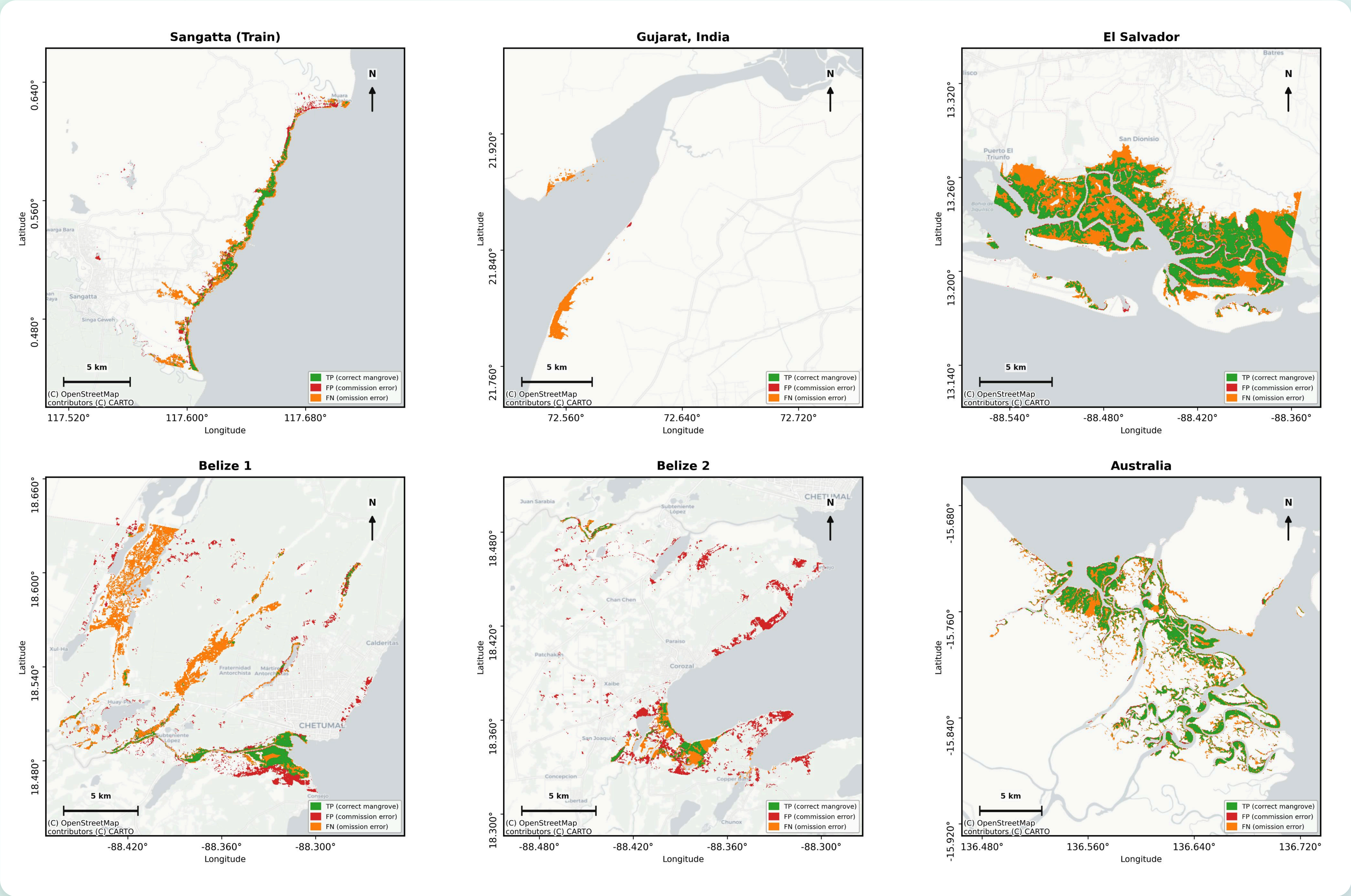
1 **Generalises**
El Salvador and Australia match or beat the training baseline (F1 0.83, 0.78).

2 **Over-predicts**
Belize keeps high recall but low precision: true mangrove found, plus commission into coastal cover

3 **Fails predictably**
Gujarat: negligible mangrove (1.87% of the zone) leaves no mode for the threshold to find.

*Sangatta = training baseline. [src: memo Table 3 - README results table]

Where the errors are: commission and omission



commission-omission error map, all six scenes. [src: NB02+NB03 - commission_omission_panel.png]

El Salvador and Australia stay clean

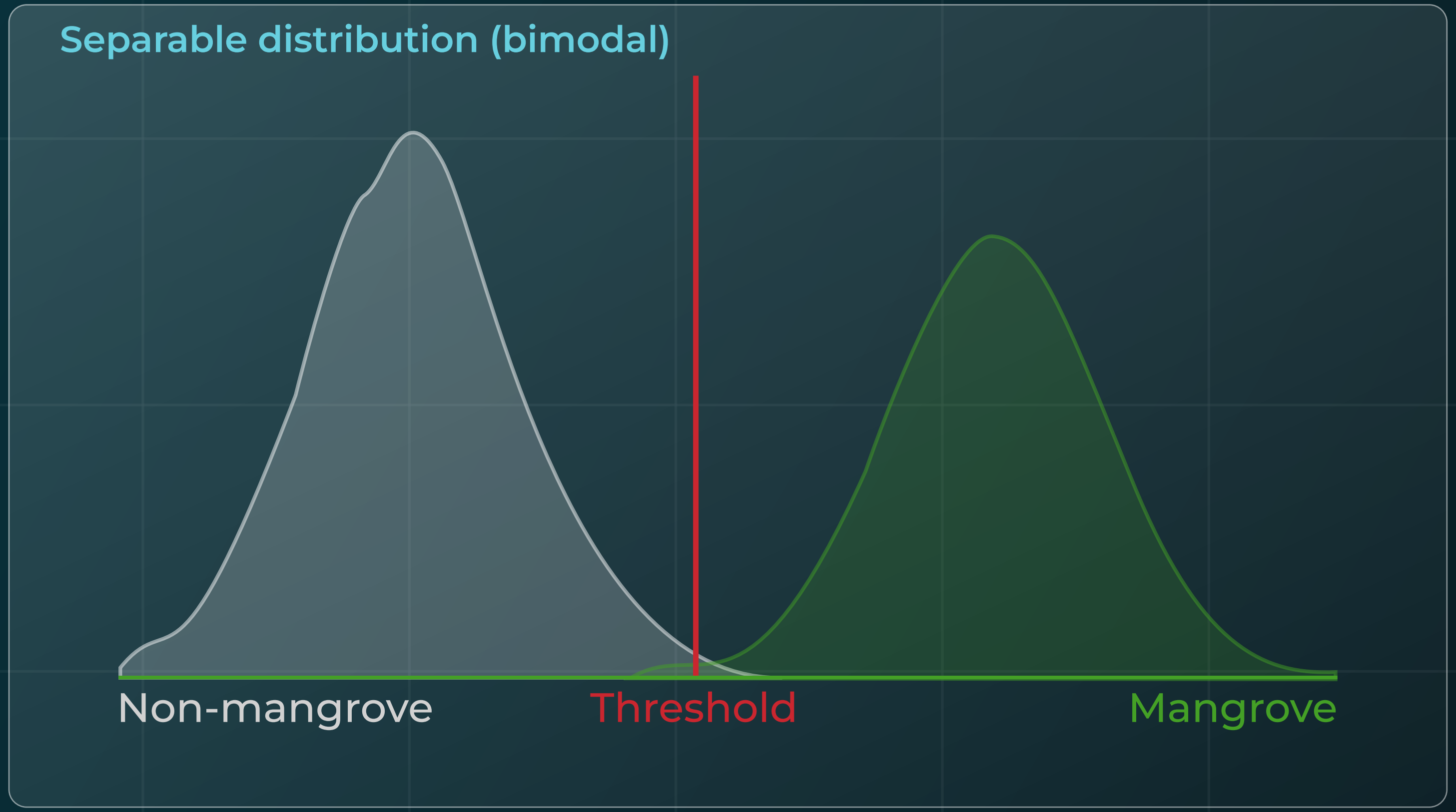
El Salvador stays in the delta (precision 0.943); Australia's inland pixels are genuine tidal-channel mangrove that GMW v3 confirms.

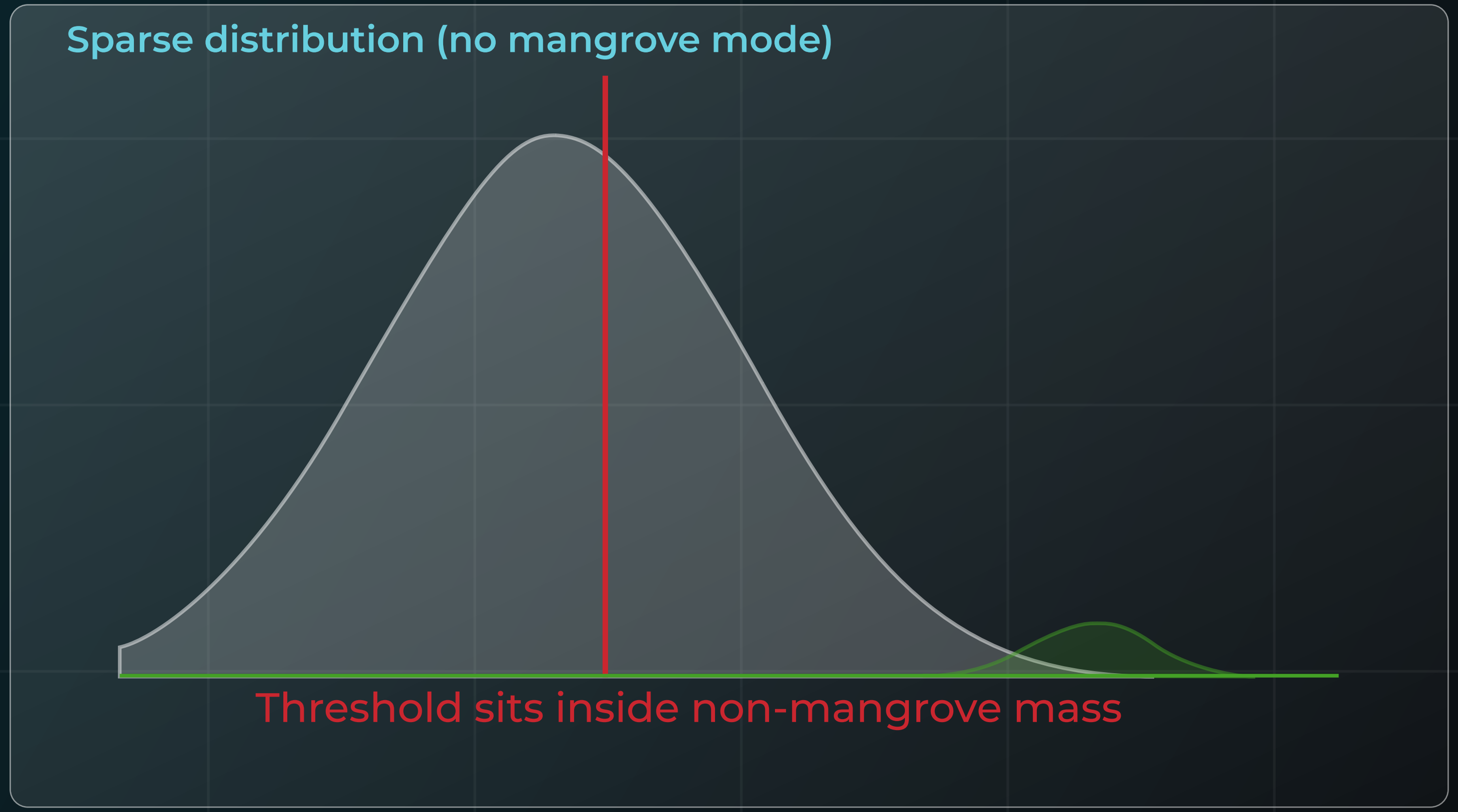
Belize over-predicts inland

The candidate zone grows from every MNDWI water pixel (creeks, rivers, lagoons), and the nine features are purely spectral, with no salinity or tidal context. Freshwater and riparian vegetation therefore leak in as commission, consistent with Munawaroh (2025).

TP FP (commision) FN (ommision)

The success/failure boundary is diagnosable





OUTCOME	SITES	DIAGNOSTIC SIGNAL
Works	El Salvador & Australia	clear bimodal MVI/NDMI
Partial	Belize 1 & 2	separable, but wet surroundings blur contrast
Fails	Gujarat, India	no mangrove detected (1.87% cover)

No field data needed: the target index distribution must be at least weakly bimodal (Nie 2026).

Honest limits, and a field-data-free path forward

LIMITATIONS

- **Pseudo-label ceiling:** CV-F1 0.962 far exceeds GMW-F1 0.615; tuning cannot close the gap.
- **Sparsity breaks the threshold:** complete failure at Gujarat (1.87% cover).
- **Temporal mismatch:** GMW v3 is 2020, scenes are 2025; change is scored as error.
- **Purely spectral:** no salinity or tidal context; REIP needs hyperspectral bands.

FUTURE WORK

1. **Blue-carbon extension**
   GEDI L4A biomass fusion, already confirmed available for the Sangatta scene.
2. **Candidate-zone refinement**
   Tidally connected water anchor, DEM-based tidal masking, and connectivity filtering.
3. **Matched validation**
   Field reference and temporally matched data to resolve the 2020-to-2025 gap.

A reference-free monitoring capability for data-scarce coasts

Hyperspectral adaptive thresholds deliver zero-shot mangrove extent without field samples, plus an interpretable account of where the method holds. It is a practical foundation for blue-carbon and biodiversity monitoring in regions that lack local training data.

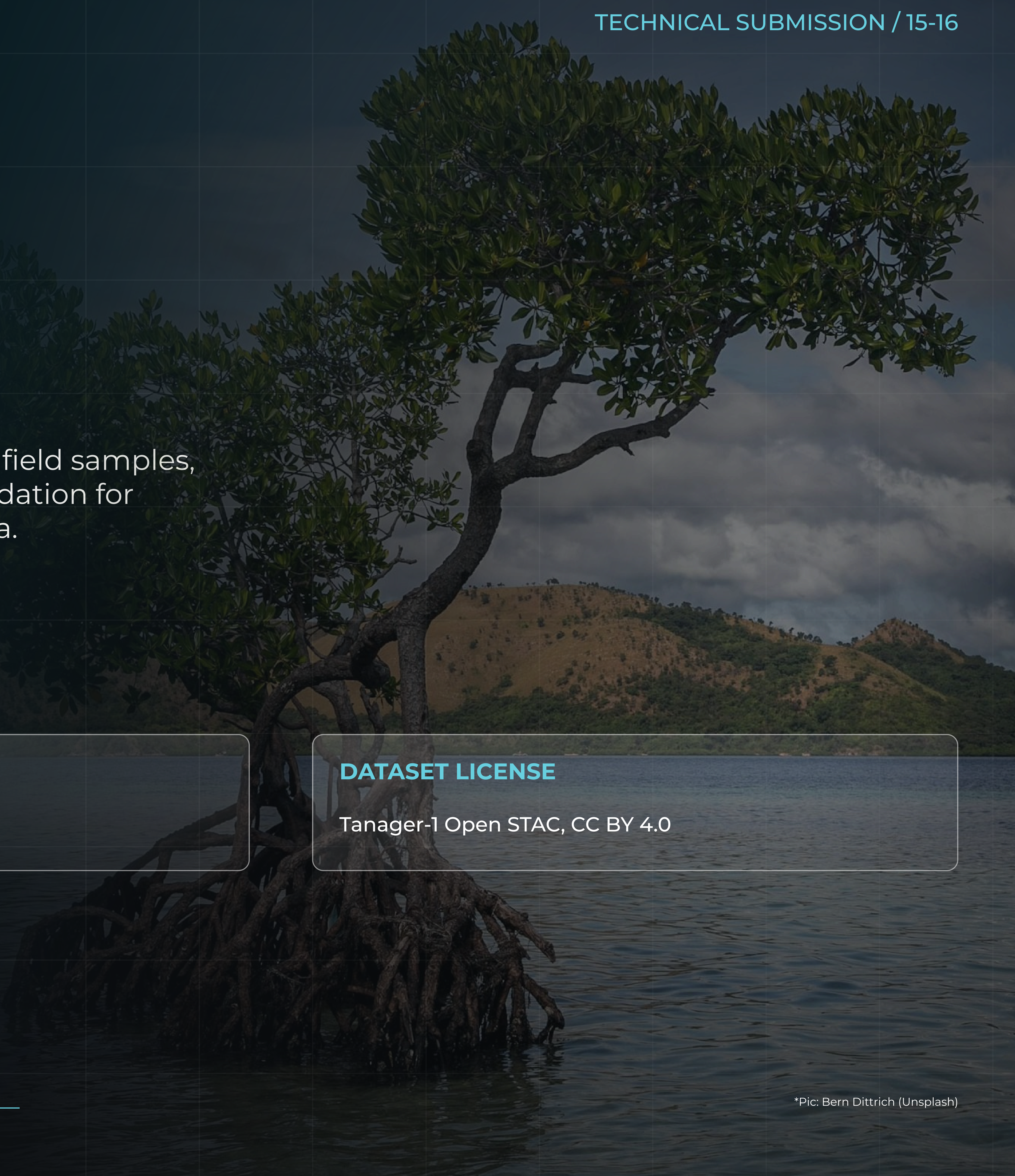
CODE (MIT)

github.com/mwahyur46/ tanager-mangrove-mapping

DATA (ZENODO)

doi.org/10.5281/ zenodo.21971757

DATASET LICENSE



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*Pic: Bern Dittrich (Unsplash)

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Full APA references with DOIs are provided in the technical memo and the project repository.